

- Q.20 Discuss any four applications of Nanotechnology in field of Medical science. (CO2)
- Q.21 Explain in brief crystallinity and reactivity property of Nanomaterials. (CO1)
- Q.22 Explain in brief about Nanocomposites. (Co1)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x8=16)
- Q.23 Explain in detail role of Nano technology in waste water treatment. (CO2)
- Q.24 Describe in detail impact of Nano Technology on environment. (CO4)
- Q.25 What is Nano Science? Discuss in detail challenges of Nanotechnology. (CO1)

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6th Sem / Chemical

Subject : Introduction to Nanotechnology

Time : 3 Hrs.

M.M. : 60

SECTION-A

- Note:** Multiple choice questions. All questions are compulsory (6x1=6)
- Q.1 Which of the following is not a Nanomaterial? (CO1)
- a) Quantum dots b) Carbon Nanotubes
 - c) Microchips d) Fullerenes
- Q.2 Nano clay is primarily composed of: (CO1)
- a) Silicon carbide b) Layered silicates
 - c) Carbon Nanotubes d) Graphene
- Q.3 Nanomaterials are often used in catalysis due to : (CO1)
- a) High optical transparency
 - b) Large surface area and active sites
 - c) Low cost
 - d) Solubility

Q.4 Use of Nanomaterials in waste water treatment is primarily due to their: (CO2)

- a) Unique physical and chemical properties
- b) High cost
- c) Lack of Porosity
- d) Radioactive nature

Q.5 X-ray diffraction (XRD) helps determine (CO3)

- a) Surface charge b) Crystal structure
- c) Electrical resistance d) Biodegradability

Q.6 Nanotechnology can help reduce environmental pollution through: (CO4)

- a) Nano particle based filtration and remediation
- b) Generating more waste
- c) Fossil fuel burning
- d) Radioactive elements

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Name any one Ceramic Nanomaterial. (CO1)

Q.8 Define Nano wires. (CO1)

Q.9 Expand FTIR. (CO3)

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Q.10 Name any one synthetic polymer Nanomaterial. (CO1)

Q.11 Carbon Nanotubes are used as conductive and semi conductive materials.(True/False) (CO1)

Q.12 Define Nano technology. (Co1)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Discuss any four social impact of Nanotechnology. (CO4)

Q.14 Explain in brief SEM characterization technique. (CO3)

Q.15 Write short note on effect of Nano Technology on Human health. (CO4)

Q.16 Describe in brief XRD technique used for Nano material characterization. (CO3)

Q.17 Discuss in brief advantages and disadvantages of chemical precipitation method used for Preparation of Nano materials. (CO3)

Q.18 Explain in brief importance of Nano Technology. (CO1)

Q.19 Discuss in brief about carbon based Nanomaterials. (CO1)

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